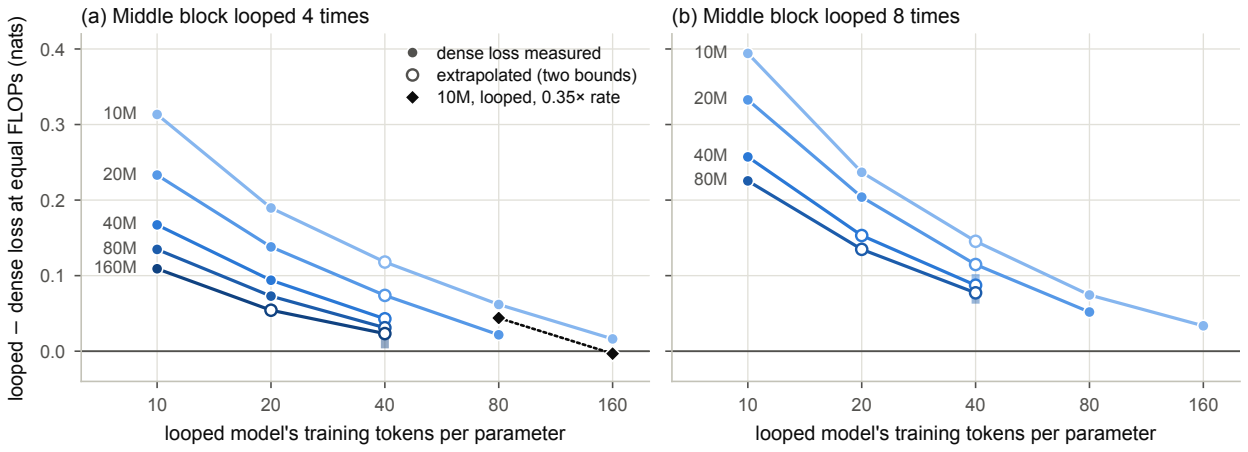
(a) Middle block looped 4 times
(b) Middle block looped 8 times
looped − dense loss at equal FLOPs (nats)
0.4
0.3
0.2
0.1
0.0
dense loss measured
extrapolated (two bounds)
10M, looped, 0.35× rate
10M
20M
40M
80M
160M
10
20
40
80
160
looped model's training tokens per parameter